

Saturated Superfluid Vacuum V Condensate Black Holes

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Draft

Abstract

This paper treats black holes in the Saturated Superfluid Vacuum (SSV) framework as coherent condensate objects rather than singular endpoints of collapse. In SSV, collapse terminates in a finite condensate state of the vacuum medium. The horizon is interpreted as an acoustic freezing surface: outward progress relative to the exterior tends to zero, but this does not imply that the object is pinned or that angular momentum must vanish. Spin is treated as a topological property of the condensate rather than as ordinary internal material rotation.

The paper also develops the claim that information is not destroyed in a singularity, because there is no singularity in the first place. Instead, horizon structure functions as the memory-bearing boundary of the condensate.

1 Introduction

This paper treats black holes in SSV as coherent condensate objects with frozen interiors, topological spin, and finite structure.

2 Black Holes as Condensates

Present the condensate picture and explain why supermassive low average density is not a problem in this framework.

3 Horizon as Acoustic-Freezing Surface

Clarify the intended meaning of freezing, local light speed, and vanishing outward escape progress.

4 Spin Without Acoustic Time

Distinguish topological winding from propagative acoustic change and explain why a frozen horizon can still carry angular momentum.

5 Information and Horizon Structure

Develop the claim that information is stored in horizon structure rather than destroyed in a singularity.

6 Discussion and Open Problems

Clarify relation to GR language, what correspondence is claimed, and what remains to be derived.

7 Conclusion

In SSV, black holes are finite coherent condensates whose horizons freeze outward progress but do not erase topology, motion, or information-bearing boundary structure.